

Mohamed Gadelhak

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Summary

I am a final-year Bachelor of Engineering (Honours) student at the University of Southern Queensland, majoring in Computer Systems Engineering, with a strong interest in electrical systems, smart infrastructure, and energy efficiency. My academic and project experiences have equipped me with a solid foundation in control systems, automation, and power systems, aligning closely with Downer's mission to design, build, and maintain the infrastructure that powers communities. Through my placement with AECOM's Buildings & Places team on a Brisbane 2032 Olympic Games project, I gained practical exposure to building services, ICT/AV systems, and sustainable design — strengthening my understanding of integrated engineering solutions across the energy and utilities sectors. As a student representative and active member of engineering initiatives, I have developed strong teamwork, communication, and leadership skills that help me thrive in collaborative, multidisciplinary environments. I am passionate about applying engineering principles to deliver safe, reliable, and sustainable energy solutions.

Technical & Professional Skills

- **Systems Analysis:** I had worked on complex circuitry in my studies for electrical circuits, where I had to use a multimeter to fault find where the circuit is shorting out thus, preventing the Lidar sensor to work/operate.
- **Project and Time Management:** Delivered an automated soccer playing robotics project, programming 4 robots with offensive and defensive roles. The project was delivered successfully by the deadline, using colour detection for the robots to track the football.
- **Data Analytics/Problem Solving:** Developed a supervised machine learning model to analyse insurance data, helping clients determine customer charges based on variables such as smoking status and number of dependents.
- **Programming:** Programmed a lawn mower camera using Python and computer vision to autonomously detect grass types and avoid obstacles based on colour segmentation.
- **Networking and Cyber Security:** I used Wireshark software to analyse and capture network packets to assist a client to detect an issue of tracking a suspicious network activity and managed to detect the user who did it by finding their IP address.
- **Teamwork:** Collaborated on a wind turbine design project, coordinating task distribution and contributing to the team's top-performing design in class.
- **Leadership/Community Involvement:** Vice President and Co-Secretary of the UniSQ Engineering Club (2024); active student member of Engineers Australia (EA) and the Australian Computer Society (ACS).
- **Presenting:** Presented the advantages/disadvantages of using co-routines in Kotlin programming languages.
- **Document Management:** created documentation, pseudocode, flowcharts and operation diagrams for the above-mentioned projects.
- **Scripting:** Competent use of Python and JavaScript, I have sound knowledge in C++ and C, I have intentions and ability to learn and master a range of programming languages.
- **Resilience and Adaptability:** Independently attended the ACS Education, Careers and Technology Conference, navigating new environments and building professional networks.

Education

- **Bachelor of Engineering (Honours) (Computer Systems Engineering)**, University of Southern Queensland (current, commenced July 2022)
- **Certificate I in Financial Services**, The Smith Family, (2021)
- **Certificate II in Information Digital Media & Technology (IDMT)**, Harristown SHS (2021)
- **Computer Programming (JavaScript, HTML, C++)**, CoderDojo Foundation (2019)
- **QCE and ATAR**, Harristown SHS, Toowoomba, Queensland (Completed Year 12 2021)

Membership and Licences

- **Engineers Australia** Student Member
- **Australian Computer Society** Associate Member

- **International Computer Driving Licence (ICDL)**

Work and Work Experience

Imagine Realities (Gold Coast) (Game developer for AR applications)

(January 2025 – July 2025, Work experience)

Duties:

- I worked with a development team to code a C++ script for a learn to tackle game using Unity game studio development platform.
- I used Orbec sensor to map out and detect human body movement, while translating it into actions to the game character.
- Explored how to use package manger in Unity as well as import and edit of in game character, animations and actions by script when certain situations occur.
- I worked with multiple teams from various displace and background to deliver a sports shoes tracking advertisement game by allowing clients to stand in front of a screen and the camera will detect their foot and apply a selected model of shoes on it.
- Developed better client management and adhered to software requirements along the development cycle incorporating agility and preserving the quality of the deliverables while adhering to project deadlines.
- Learnt about the working environment within start-up technology companies and the importance of keeping positive client relationships especially in such early stages of the business.
- Conducted one on one training and development sessions with the CTO of the company.

Toowoomba Regional Council (ICT Service Desk officer)

(September 2025 – November 2025, Work experience)

Duties:

- Worked with a team of officers to receive calls and document incidents on Cherwell Client Management Software.
- Assisted in delivering and closing out several service desk level 1 and 2 tasks by attempting to troubleshoot or fix customer's issues remotely or conducting site visits to preform equipment replacement.
- Journalised over 150 entries for hardware and service plans purchases to assist the financial department with keeping records to charge relevant departments/teams based on what equipment they ordered/used.
- Conducted over 5 warranty repair claims with DELL computers by using their business hub portal.
- Assisted in the delivery of windows 11 updates program and replacement hardware program across the entire organisation.
- Prepared a report on risk management, identifying possible short comes within the helpdesk team to improve the quality of customer's experience and satisfaction levels while focusing spirally on department mangers reactions.

AECOM Australia Pty Limited (Fortitude Valley) (PVR Principal Consultant)

(October 2023 – February 2024, work experience)

Project: Brisbane 2032 Project Validation Report (PVR)

Reference Design: Major Venues Project (Toowoomba)

Client: Department of Energy & Public Works (DEPW)

- Observed architectural and engineering team develop the reference design, with the opportunity to challenge design progression during student feedback sessions. Discuss with stakeholders including Queensland Police Service, Department of Tourism and Sport and Queensland Government (Olympics Lead) their needs for the project.
- Reviewed design and engineering documentation with the Principal Consultant team to understand why decisions were made for the project.
- Exposed to crowd modelling software, the use of Grasshopper and Lumion design tools, and how the engineers document architectural drawings for reference design projects.

Referees

Ms Kara Lansdell

Principal Project Officer, Minor Venues Program

Games Venue & Legacy Delivery Authority

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Note that further referees can be provided upon request